Decoding Data Science(DDS) Academy

AI Application Building Challenge

For students (instructor review required)

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Day 1 Idea Submission Template

Project Title:

BlackRoute – AI agents that find weak spots in supply chains before they break

Concept Summary:

(Briefly describe your idea, including its purpose and the problem it aims to solve.)

When one port or shipping route is blocked, shipments get delayed, warehouses run low and customers wait. Most companies only find out when it is already too late.

BlackRoute is an AI app that warns you early and helps you fix the problem. It finds the weakest routes in your supply chain (the "black routes"), watches news, weather and shipping alerts, and uses a team of AI agents to plan a fix. Every plan is tested in a simulator, and a human makes the final choice.

AI finds the problem. The simulator checks the plan. A human decides.

Target Audience:

(Who will benefit from your project? Describe their needs.)

- Logistics and supply chain planners at small and medium companies, like distributors, freight forwarders and warehouse operators.
- Shop, factory and warehouse managers who need their stock to arrive on time.

They need early warnings, to know who will run out first, and clear plans they can trust. Big companies have expensive tools for this. Smaller ones usually don't.

Key Features:

(List the main features or functionalities of your application.)

1. **Black Route Radar:** finds the weakest routes before anything goes wrong.
2. **Live alerts:** an AI agent reads news, weather and shipping alerts and flags real risks, with sources.

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3. **Team of 5 AI agents:** each agent has one job (watch, check impact, plan, challenge, write actions). You can see what each one is doing.
4. **Recovery simulator:** tests the cheapest, fastest and balanced plans in hundreds of possible futures, and shows the cost of waiting.
5. **Red Team check:** one agent tries to break the plan before a human sees it.
6. **Human approval:** approve, reject or change the plan. The app then writes the emails and reports for you to send.
7. **3D map:** shows the problem spreading, and then the recovery.

Technical Approach:

(How do you plan to implement your idea? Mention technologies or methods.)

Path: Code-First

- **Claude Code:** my AI coding partner that builds the whole app.
- **Web app:** React for the dashboard and approval screen, Three.js for the 3D map.
- **AI agents:** free AI APIs (like Google Gemini) with tool calling, so each agent can use the simulator and live data.
- **Simulator:** plays out what happens hour by hour and tests each plan in hundreds of possible futures.
- **Supabase:** database for alerts, agent messages and decisions.
- **Free data:** GDELT for news and Open-Meteo for weather.
- **Vercel:** puts the app online.

Expected Challenges:

(What potential obstacles do you foresee, and how will you address them?)

- **AI making things up:** numbers only come from the simulator, and every alert shows its source.
- **Too much useless news:** filter by relevance, and a human confirms big alerts.
- **Agents getting confused:** each agent has one clear job and works in a fixed order.
- **No real company data:** use real ports and routes with a made-up company.
- **Only 8 days:** build one full working demo first, then add extras.
- **Free API limits:** use a fast model for simple jobs, save results, and keep a backup demo mode so the demo always works.

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Submission Format:

(Outline how your submission should be presented: documents, slides, etc.)

- **Day 1:** this document and a concept image.
- **Checkpoints:** short progress notes with screenshots or short videos.
- **Final:** live app link, a 2–3 minute demo video, short slides and a LinkedIn post.

Expected Outcome:

(What do you aim to achieve by the end of the challenge?)

A live, working app where anyone can:

- see the weakest routes on a 3D map
- watch the AI agents catch a problem (for example, a Red Sea shipping delay that slows goods going to UAE warehouses)
- compare recovery plans and approve one
- watch the 3D map show the recovery.

Additional Notes (Optional):

- The whole app is built with Claude Code as my AI coding partner.
- The company and warehouses in the demo are made up. The ports and routes are real.

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